

The Final Presentation

Final Listing Price Prediction for Private Used Car Sellers



Group 2

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1. Introduction

News

How much profit do car dealers make on new and used cars? Here's the perception versus reality

- Huge disparity between how much buyers think dealers make and the reality
- Some consumers think dealers are making more than 75 per cent profit per car
- What Car? survey shows average consumer thinks dealers make 10-20 per cent per unit
- Dealers reveal that they really make about seven per cent on new cars

The used car market is a lot stronger with profit margins for dealers around 12 to 15 per cent.

David Kendrick, partner and accountancy UHY Hacker Young, said the figures dealers quoted to Car Dealer are accurate.

He said: 'There is a huge misconception as to how much retailers make on vehicle sales.

'We take on a large number of graduates every year and one of our questions is how much the average dealer makes on a £15k vehicle.

'Answers range from £10k down to those who understand it may be minimal to a loss.

'The issue is that there is such a huge range and misunderstanding. The substantial multi-million pound dealerships certainly don't help that perception. I am not sure how this will ever change.'

Kendrick said the average most dealers make on new and used cars is around £1,000 a car.

Dealers benefiting from the difference between the selling price and the buying price of a used car

➡ **Sellers** have difficulty measuring the price they can actually obtain..😭

1. Introduction

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Brief

The Outlook for the European Used Car Market

Growth opportunities may be closer than they appear.

By Roch Baranowski, Eric Zayer, Klaus Stricker, and Ingo Stein

10 min read



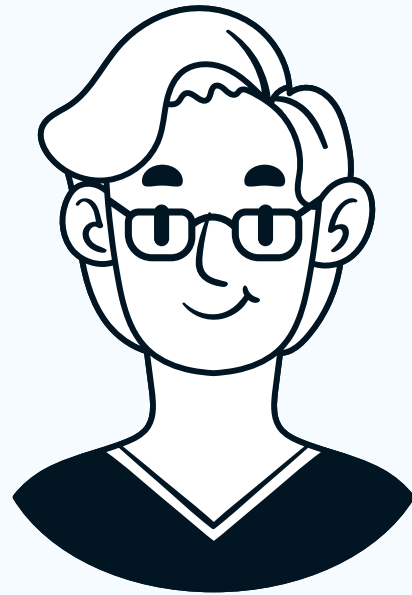
With a total volume of 429 billion euros in 2021, the market for used cars in Europe is similar to the market for new cars. Of the 32 million used cars sold in Europe in 2021, some 44% were sold privately in consumer-to-consumer (C2C) transactions, while 56% were sold by professional retailers, which tend to focus on higher-class, younger cars. Furthermore, used car margins typically beat new car margins for dealers. Overall, the used car market is growing healthily, with a CAGR of 7% between 2015 and 2021.

Private Transactions are not easy to investigate with official statistics

➡ **Buyers** also have difficulty determining whether the sale posted on the used car site is a false sale or not..😭

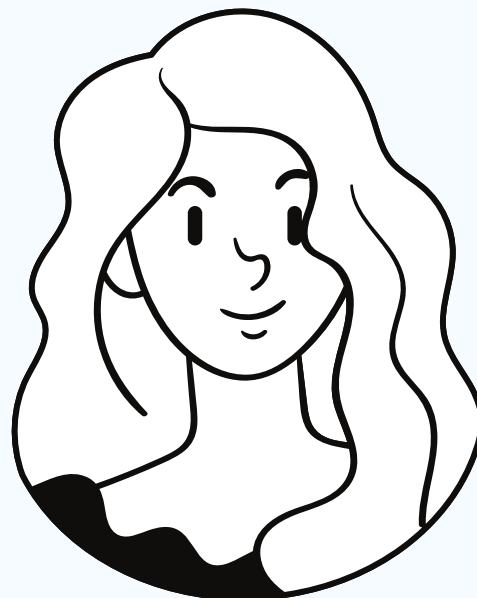
4

2. Target Audience Analysis - **Seller** & **Buyer**



Daniel Brühl - **Seller**

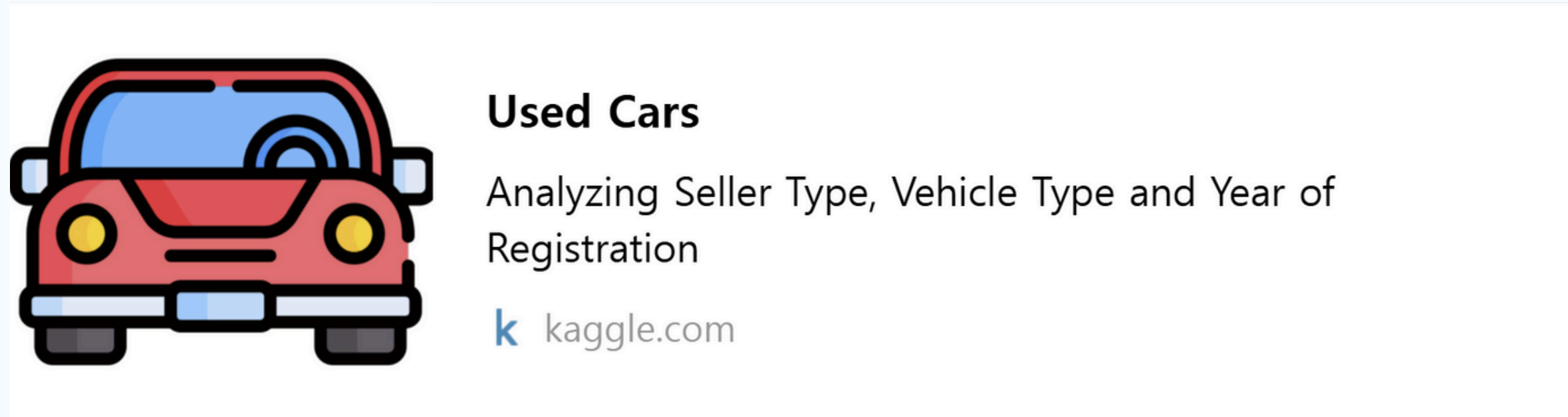
- Trying to sell my original car in order to buy a new car!
- But I'm afraid of the dealer fraud....
- Can't I just **sell this privately** with "Appropriate Price"?



Maria Stein - **Buyer**

- Want to buy used car cheaper because I'm out of budget.
- Then I have to find a used car in private sale, without a dealer...
- How much will be "the common selling price" of my dream car?

3. Our Dataset



21 features

X

233531 data samples



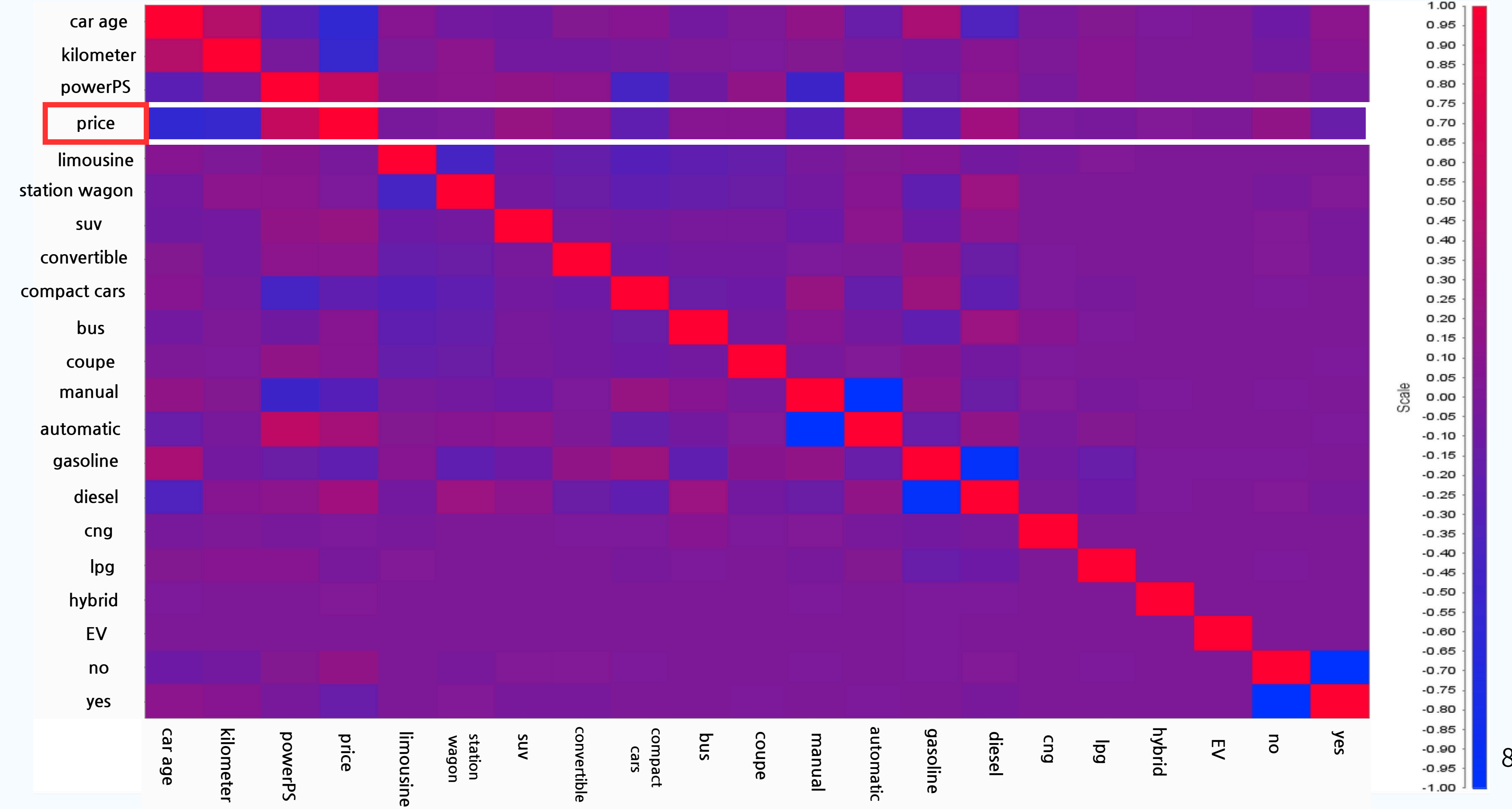
dateCrawled, name, seller,
offerType, price, abtest, vehicleType,
yearOfRegistration, gearbox etc.

Source of data : ebay or other sites where cars can be posted.

3. Our Dataset

car_brand	car_model	car_age	kilometer	vehicleType	gearbox	powerPS	fuelType	notRepairedDamage	price
volkswagen	golf	19	150000	limousine	manual	75	gasoline	no	600
volkswagen	golf	17.58333	150000	station wagon	manual	90	diesel	no	1459
volkswagen	passat	7.666667	150000	station wagon	manual	170	diesel	no	5890
volkswagen	golf	9.75	150000	limousine	manual	80	gasoline	no	4000
volkswagen	touareg	10.91667	150000	suv	automatic	224	diesel	no	11900
volkswagen	beetle	12.66667	150000	convertible	manual	75	gasoline	no	5800
volkswagen	fox	5	80000	compact cars	manual	60	gasoline	no	7800
volkswagen	golf	4	60000	limousine	manual	211	gasoline	no	19200
volkswagen	golf	15.83333	150000	limousine	manual	75	gasoline	no	1200
volkswagen	golf	32.16667	150000	limousine	manual	75	gasoline	no	3799
volkswagen	golf	6.166667	90000	limousine	manual	105	diesel	no	8990
volkswagen	golf	21.75	150000	convertible	manual	90	gasoline	no	1750
volkswagen	passat	7.5	150000	station wagon	manual	105	diesel	no	5100
volkswagen	passat	10.58333	150000	limousine	manual	140	diesel	no	6700
volkswagen	golf	5.833333	50000	station wagon	manual	80	gasoline	no	9899
volkswagen	polo	12.08333	125000	limousine	automatic	75	gasoline	no	4900
volkswagen	polo	15.83333	150000	station wagon	manual	75	gasoline	no	1690
volkswagen	golf	7.666667	150000	limousine	automatic	105	diesel	no	8499
volkswagen	polo	11.91667	125000	compact cars	manual	72	gasoline	no	3000
volkswagen	sharan	10	150000	bus	manual	140	diesel	no	6350
volkswagen	golf	2.833333	40000	convertible	manual	211	gasoline	no	20450
volkswagen	golf	4.416667	125000	limousine	automatic	170	diesel	no	15900

3. Our Dataset



4. Data Preprocessing Recap

- Outliers (Too big or small value)
- Ridiculous Values
- Duplicated Data

*_Mercedes-Benz_W220_S_400_CDI_Langvers._<>_162_TKM_>_]	9999			0 s_klasse	150000	mercedes_benz
*_Mercedes-Benz_W220_S_400_CDI_Langvers._<>_162_TKM_>_]	9999			0 s_klasse	150000	mercedes_benz
*_Mercedes-Benz_W220_S_400_CDI_Langvers._<>_162_TKM_>_]	9999			0 s_klasse	150000	mercedes_benz
*_Mercedes-Benz_W220_S_400_CDI_Langvers._<>_162_TKM_>_]	9999			0 s_klasse	150000	mercedes_benz
*_Mercedes-Benz_W220_S_400_CDI_Langvers._<>_162_TKM_>_]	9999			0 s_klasse	150000	mercedes_benz
*_Mercedes-Benz_W220_S_400_CDI_Langvers._<>_162_TKM_>_]						mercedes_benz
*_Mercedes-Benz_W220_S_400_CDI_Langvers._<>_162_TKM_>_]						mercedes_benz

#	R... ↓	count ↓ <i>Number (integer)</i>
155	9999	27
154	9996	1
153	9450	1
152	9229	1
151	9000	5
150	8888	2
149	8500	1
148	8455	1
147	8200	1
146	8000	2
145	7800	1
144	7777	1
143	7500	2
142	7100	1

#	R... ↓	count <i>Number (integer)</i>
89	2018	3
88	2017	7
87	2016	283
86	2015	2027
85	2014	3143
84	2013	3817
83	2012	5479
82	2011	6594
81	2010	6584
80	2009	7905
79	2008	8220
78	2007	8841
77	2006	9958
76	2005	9536

Dialog - 4:94 - Row Filter (PS)

File

Filter Criteria | Flow Variables | Job Manager Selection | Memory Policy

☒ Include rows by attribute value
☐ Exclude rows by attribute value
☐ Include rows by number
☐ Exclude rows by number
☐ Include rows by RowID
☐ Exclude rows by RowID

Column value matching

Column to test:

☐ filter based on collection elements

Matching criteria

☐ use pattern matching

☐ case sensitive match ☐ contains wild cards
☐ regular expression

☒ use range checking

lower bound:
upper bound:

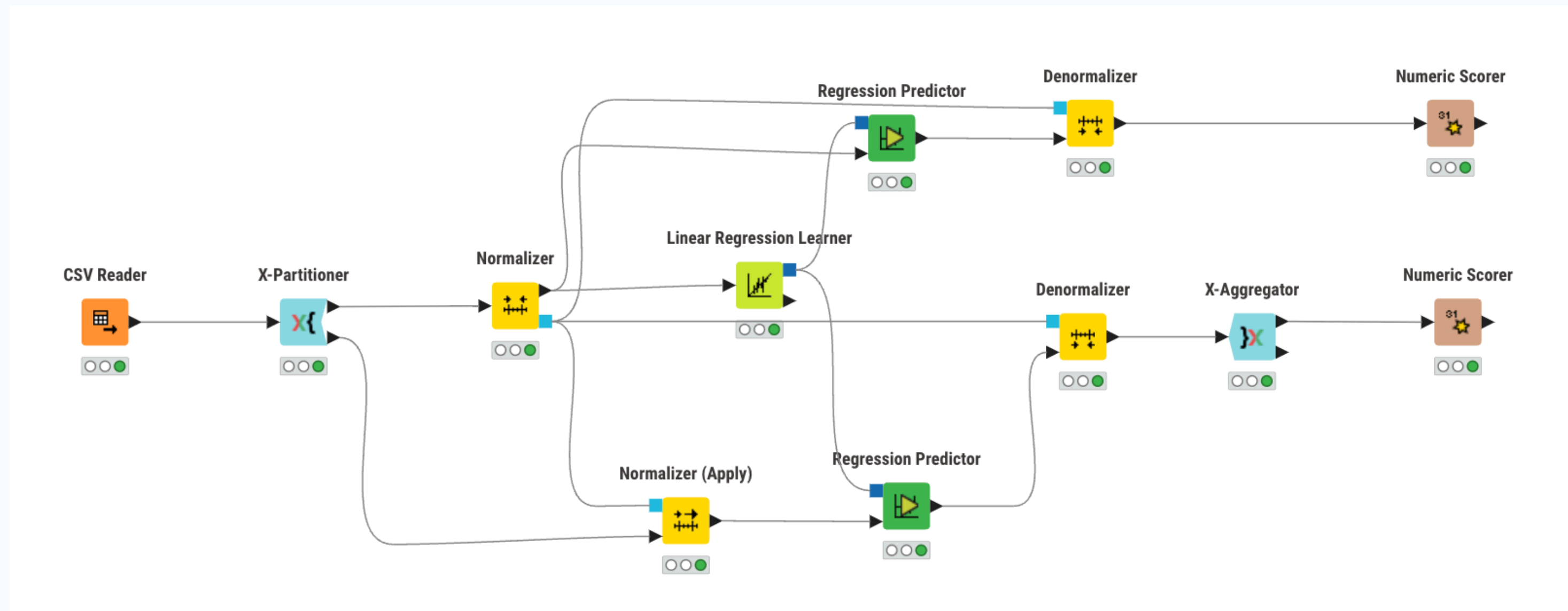
☐ only missing values match

OK Apply Cancel ?

5. Model - Our Evaluation metrics

Evaluation metric	Advantage	Disadvantage	Why Selected?
R²	Evaluate how much the model describes price volatility & explanatory	Sensitive to outliers	Suitable for assessment of explanatory power
MAPE	Intuitive with prediction error as a percentage	Small prices may distort the result	Intuitively assess actual predictive performance
RMSE	More accurate reflection of prediction errors in high-priced vehicles	Sensitive to outliers	Useful for performance evaluation of high-priced vehicle predictions

5. Model Works - Multiple Linear Regression (MLR)



5. Model Works - Multiple Linear Regression (MLR)

Nominal values for MLR : Label encoding

Q. Why not One-hot Encoding?

A. Experienced the **Curse of Dimensionality**...🙄

Fixed) Linear correlation > 0.5 : car_age, kilometer, powerPS, car_model

vehicleType(1), gearbox(2), fuelType(3), notRepairedDamage(4)

CASE	x	1, 3, 4	2, 3, 4	1, 2, 3, 4
R^2	0.63506842	0.648601815	0.656494454	0.655735274
mean absolute error	3174.356195	3132.950155	3118.916731	3124.028734
mean squared error	25281779.31	24440843.29	23588280.12	23578420.14
root mean squared error	5028.098976	4943.768127	4856.776721	4855.761541
mean signed difference	-5.4771E-10	-4.02865E-09	4.30216E-10	4.01539E-10
mean absolute percentage error	0.906901314	0.903172851	0.910689249	0.899603956
adjusted R^2	0.63506842	0.648601815	0.656494454	0.655735274

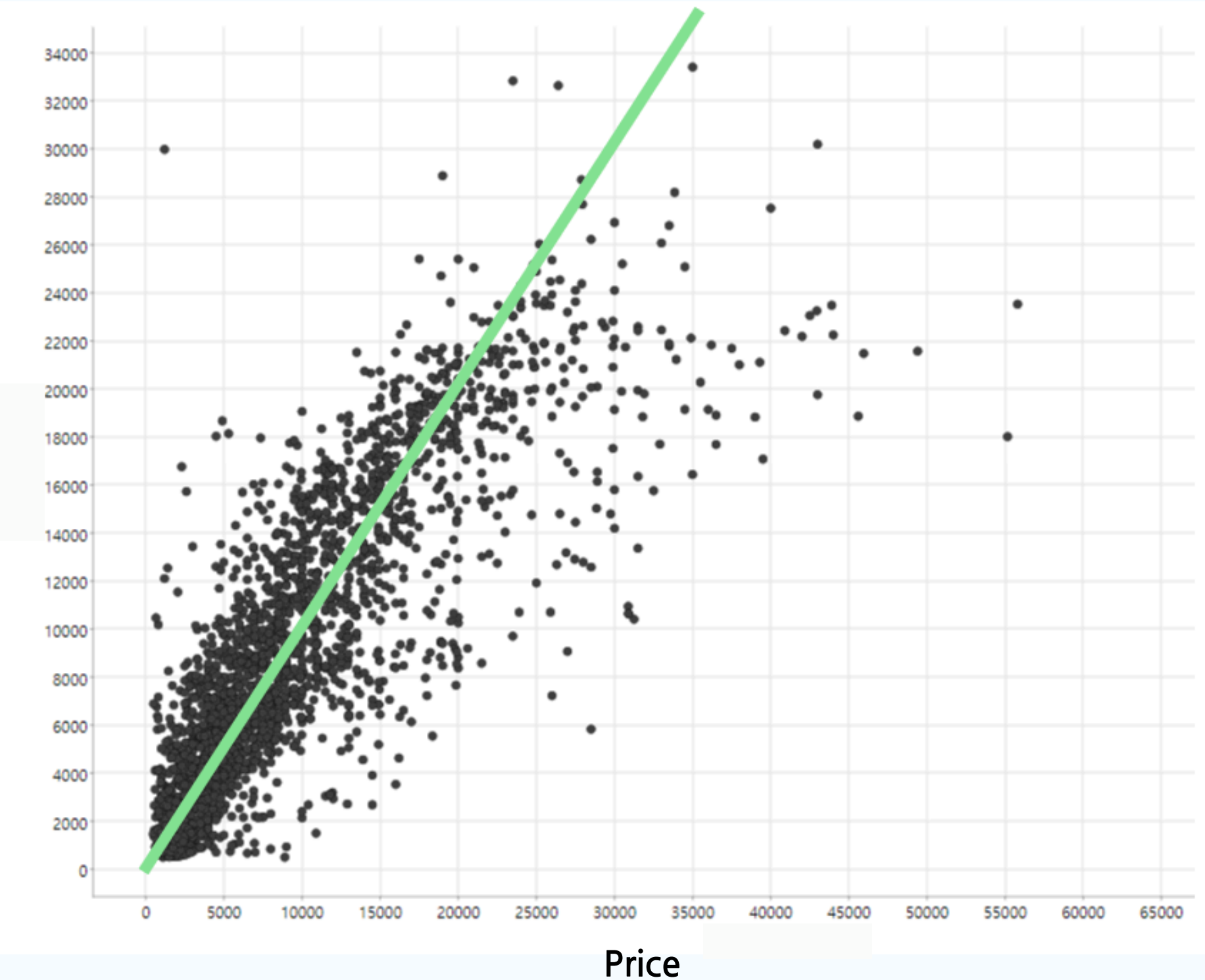
Using selected features

Using all features

5. Model Works - Multiple Linear Regression (MLR)

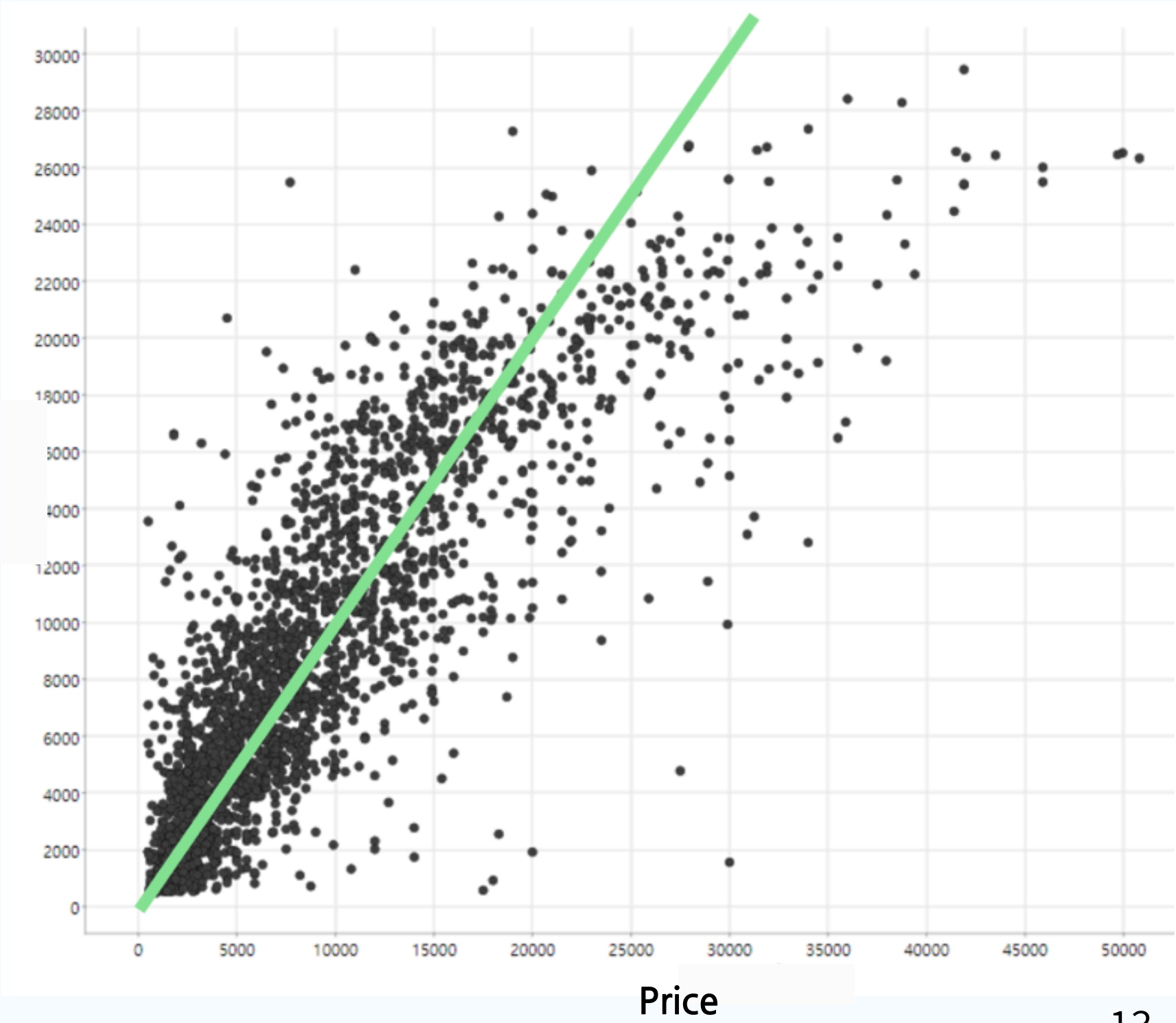
Prediction Price

Using selected features

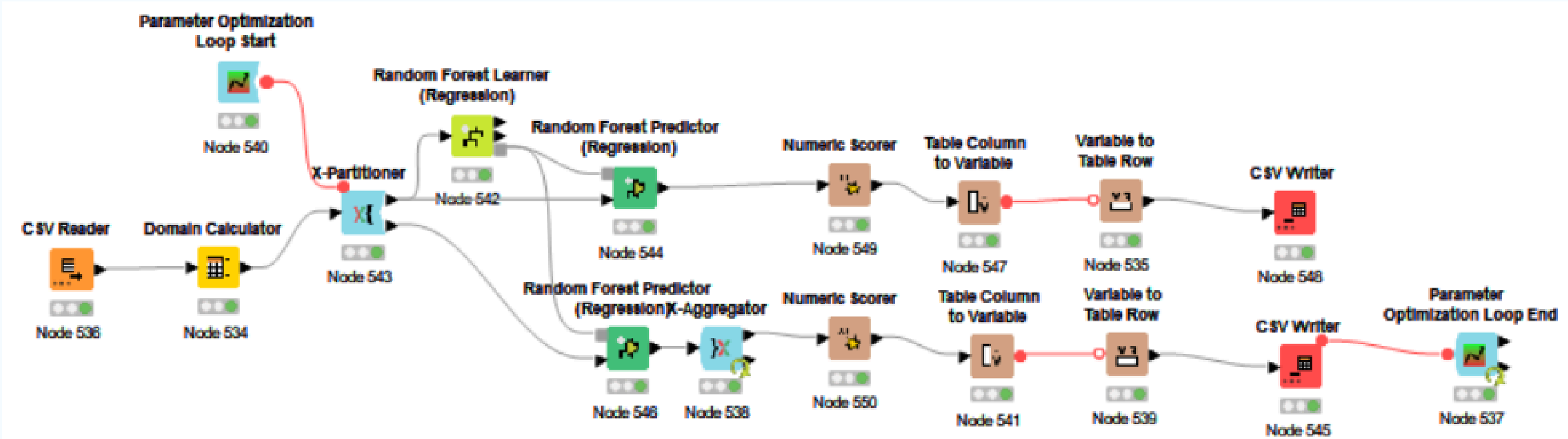


Prediction Price

✓ Using all features



5. Model Works - Random Forest



5. Model Works - Random Forest

- ★ **Limit number of levels (tree_depth)** : Number of tree levels to be learned
- ★ **Minimum child node size (min_child_size)** : Minimum number of records in child nodes

	Start Value	Stop Value	Step Size	Best Param
tree_depth	3	100	5	73
min_child_size	5	30	5	5

performance	prediction (price)
R^2	0.88045
MAPE (Mean Absolute Percentage Error)	0.356
RMSE (Root Mean Squared Error)	2882.6

5. Model Works - Random Forest

	Start Value	Stop Value	Step Size	Best Param
tree_depth	50	90	1	69
min_child_size	5	6	1	5

★Optimal Hyperparameter

tree_depth : 69 / min_child_size : 5

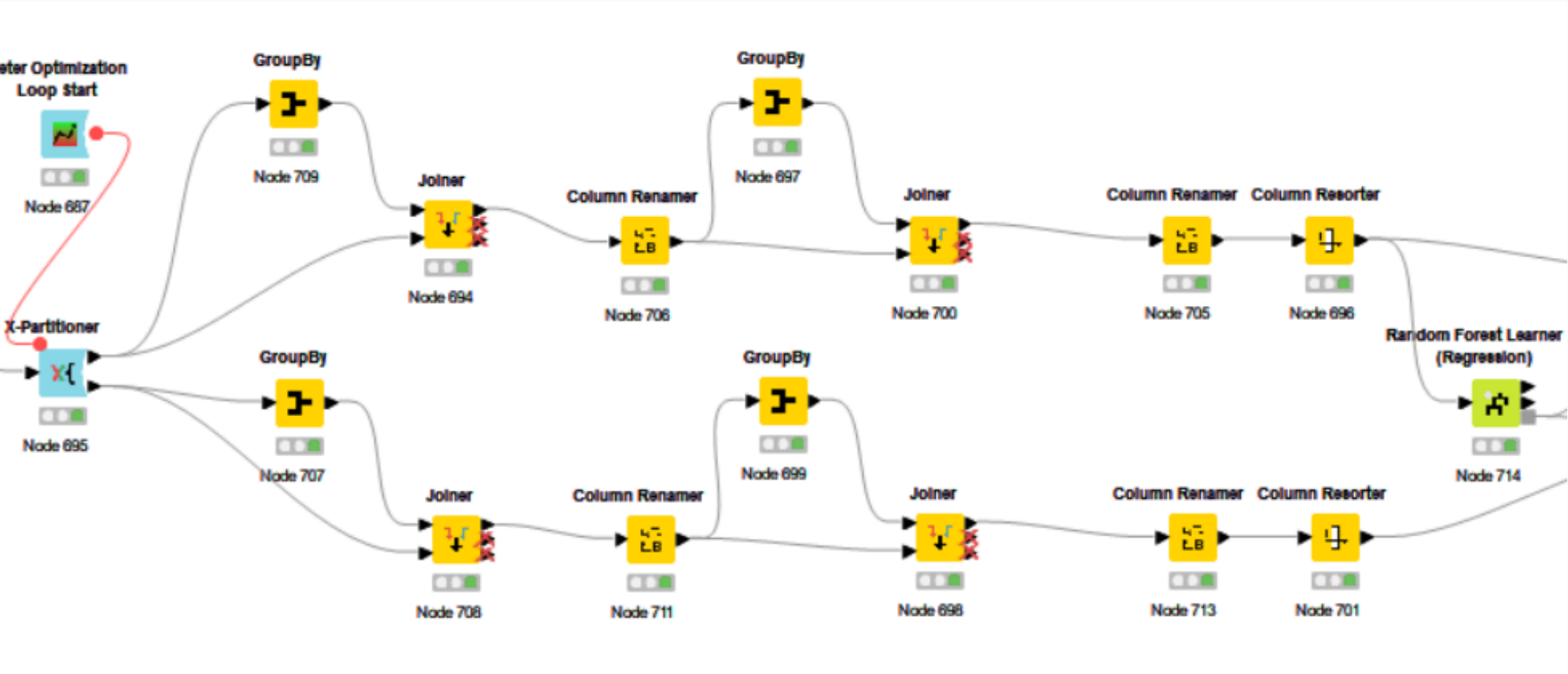
performance	prediction (price)
R^2	0.8808
MAPE (Mean Absolute Percentage Error)	0.356
RMSE (Root Mean Squared Error)	2878.83

5. Model Works - Random Forest

★Target Encoding..?

Convert brands and models to average prices

car_brand	car_model	car_brand_enc	car_model_enc	price
volkswagen	golf	6819.045426	6024.54069	600
volkswagen	golf	6819.045426	6024.54069	1459
volkswagen	passat	6819.045426	6013.214716	5890
volkswagen	golf	6819.045426	6024.54069	4000
volkswagen	touareg	6819.045426	17980.71884	11900
volkswagen	beetle	6819.045426	8735.380631	5800
volkswagen	fox	6819.045426	2997.835766	7800
volkswagen	golf	6819.045426	6024.54069	19200
volkswagen	golf	6819.045426	6024.54069	1200
volkswagen	golf	6819.045426	6024.54069	3799
volkswagen	golf	6819.045426	6024.54069	8990
volkswagen	golf	6819.045426	6024.54069	1750
volkswagen	passat	6819.045426	6013.214716	5100



in order to prevent ‘the data leakage’...

5. Model Works - Random Forest

	Start Value	Stop Value	Step Size	Best Param
tree_depth	90	110	1	100
min_child_size	5	6	1	5

★Optimal Hyperparameter

tree depth : 100 / min_child_size : 5

performance	prediction (price)
R^2	0.878
MAPE (Mean Absolute Percentage Error)	0.37
RMSE (Root Mean Squared Error)	2911.78

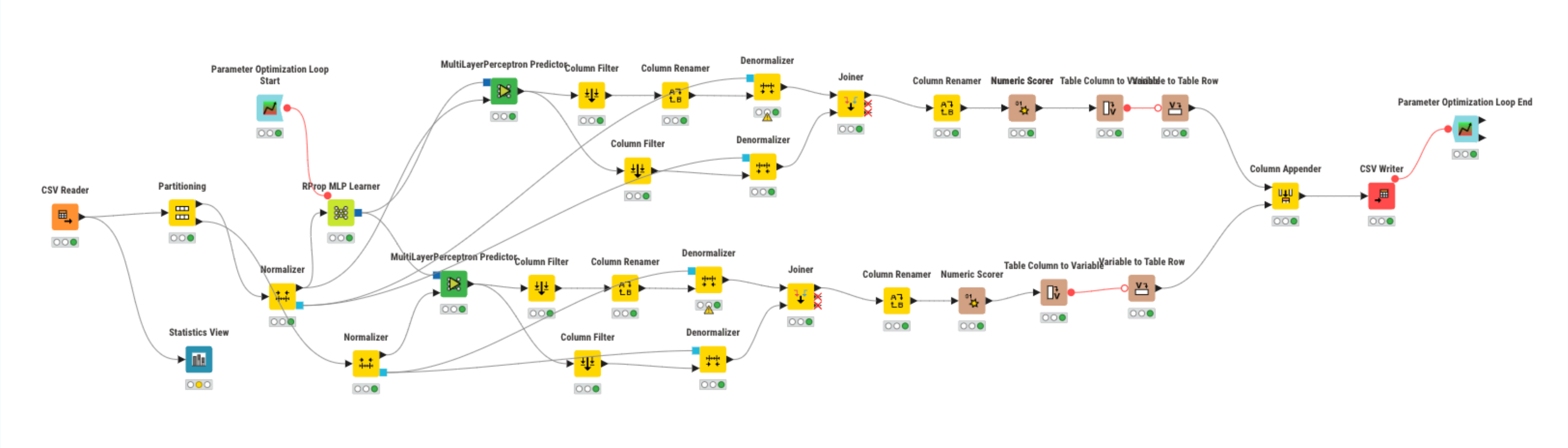
Original Dataset (Unencoded)

		Min child node size			
		5	6		
Tree depth	60	0.879586	0.876837		
	61	0.878303	0.876403		
	62	0.880456	0.876057		
	63	0.878985	0.878191		
	64	0.878234	0.877959		
	65	0.876653	0.878279		
	66	0.879733	0.878207		
	67	0.878702	0.877476		
	68	0.877936	0.876826		
	69	0.880798	0.876526		
	70	0.879721	0.876119		
	71	0.87931	0.876935		
	72	0.878709	0.877494		
	73	0.879647	0.877303		
	74	0.879002	0.875762		
	75	0.878169	0.876279		

Target Encoded Dataset

		Min child node size			
		5	6		
Tree depth	90	0.876485	0.874156		
	91	0.87578	0.875022		
	92	0.877532	0.875158		
	93	0.877572	0.875161		
	94	0.875064	0.876897		
	95	0.877696	0.87456		
	96	0.877193	0.874161		
	97	0.877141	0.874666		
	98	0.877741	0.873691		
	99	0.875669	0.874567		
	100	0.878054	0.874959		
	101	0.876756	0.874575		
	102	0.877514	0.87576		
	103	0.877466	0.874616		
	104	0.877498	0.875535		
	105	0.877946	0.874174		
	106	0.877408	0.874065		
	107	0.877069	0.876749		
	108	0.877184	0.875546		
	109	0.877206	0.874147		
	110	0.877418	0.875256		

5. Model Works - Multi-Layer Perceptron (MLP)



5. Model Works - Multi-Layer Perceptron (MLP)

performance	prediction (price)
R^2	0.672
MAPE (Mean Absolute Percentage Error)	0.404
RMSE (Root Mean Squared Error)	4659.79

5. Model Works - Multi-Layer Perceptron (MLP)

Number of hidden layers

Number of hidden neurons per layer

	2	3	4	5	6	7	8	9	10
3	0.65017	0.615928	0.614016	0.622766	0.509766	0.573377	0.370194	0.403561	0.39479
4	0.642803	0.632683	0.6247	0.610568	0.603258	0.512381	0.557201	0.355451	0.334034
5	0.639394	0.633924	0.632836	0.629655	0.606527	0.596536	0.584553	0.548112	0.511847
6	0.643445	0.641206	0.617749	0.631223	0.621891	0.615441	0.569108	0.433752	0.361475
7	0.635318	0.639291	0.652641	0.623891	0.653204	0.605677	0.604397	0.585897	0.556575
8	0.648531	0.650159	0.642627	0.631599	0.642783	0.572472	0.640443	0.532577	0.596598
9	0.650508	0.656796	0.639056	0.64895	0.649066	0.648743	0.59522	0.576442	0.603677
10	0.669919	0.648926	0.64796	0.644714	0.639783	0.649168	0.640463	0.567658	0.588777
11	0.65793	0.65408	0.658042	0.639319	0.650362	0.651778	0.621998	0.645935	0.63396
12	0.658547	0.654088	0.644217	0.644421	0.652721	0.634978	0.638008	0.618387	0.650662
13	0.658306	0.659108	0.649263	0.654403	0.659177	0.635069	0.651555	0.642547	0.632144
14	0.661775	0.653396	0.657138	0.654256	0.657063	0.636648	0.638928	0.636911	0.638158
15	0.650996	0.651512	0.657177	0.662008	0.658917	0.646684	0.650571	0.643291	0.592856
16	0.661198	0.654775	0.656722	0.655743	0.657195	0.655165	0.639675	0.638021	0.626475
17	0.66177	0.665334	0.652482	0.655764	0.656715	0.658282	0.619201	0.650585	0.648881
18	0.668007	0.656835	0.660502	0.651676	0.651214	0.658392	0.649922	0.641516	0.636192
19	0.67212	0.666316	0.654828	0.66083	0.659117	0.655186	0.652515	0.634711	0.648319
20	0.671967	0.655871	0.663365	0.657645	0.651424	0.641183	0.647533	0.645191	0.643466
21	0.662363	0.651803	0.654011	0.647677	0.65368	0.65391	0.653154	0.649486	0.627727
22	0.66328	0.656507	0.666222	0.658549	0.657493	0.651311	0.647223	0.655864	0.645305
23	0.659553	0.669976	0.655335	0.649904	0.651897	0.643876	0.643862	0.648783	0.623982
24	0.657986	0.655509	0.658985	0.663448	0.650668	0.65276	0.64579	0.652146	0.616976
25	0.659514	0.650194	0.653474	0.664067	0.657976	0.653952	0.660429	0.651348	0.656586

6. Model Results

Performance	Multiple Linear Regression	Random Forest	Multi-Layer Perceptron
R^2	0.656	0.878	0.672
MAPE (Mean Absolute Percentage Error)	0.899	0.37	0.404
RMSE (Root Mean Squared Error)	4855.76	2911.78	4659.79

7. Conclusion

Q1. Why Random Forest?

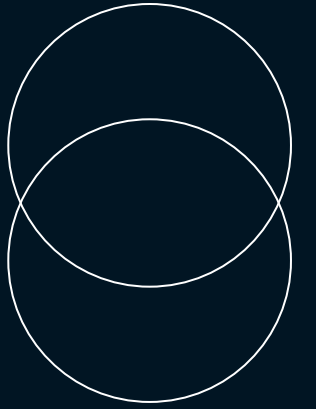
A. Since it's an ensemble model of individual decision trees...

- Good at catching **nonlinear relationships**
- Good at handling **categorical variables** well
- Can effectively reflect interactions between variables
- Automatically calculates feature importance -> **use important variables**

Q2. Why not MLP?

A. Without sufficient training data and adequate hyperparameter tuning, MLP may not capture nonlinearity well...

- Requires **learning all its characteristics** & consideration of **non-critical variables** => Can lead to a lot of unnecessary learning!



Q & A

Thank you for listening!